

Amazon S3 Object Lock

To Begin with the Lab

Summary of the Lab

In this lab, **Amazon S3 Object Lock** was used to protect objects from deletion or modification for a fixed time or indefinitely. The video explained the two retention modes, **Governance mode** and **Compliance mode**, and showed how **Legal Hold** can be used when there is no fixed expiration date. It also demonstrated that Object Lock requires **versioning** to be enabled first, and that once Object Lock is enabled, it cannot be disabled later. The lab showed how to enable Object Lock when creating a bucket or after creation from the bucket properties, then apply retention settings to uploaded objects. Finally, it verified that locked objects could not be deleted or overwritten until the protection was removed or the retention period ended.

Understanding Amazon S3 Object Lock

Amazon S3 Object Lock is a protection feature that prevents an object version from being deleted or changed for a set period of time, or indefinitely. It exists so important data can be protected from accidental deletion, malicious deletion, or unwanted edits. **Governance mode** is the softer option because users with special permissions can still bypass or change the lock, while **Compliance mode** is strict and cannot be overridden by anyone, including the root user, until the retention period ends. A **Legal Hold** is different because it has no built-in end date and must be removed manually.

Example:

A bank stores customer records that must be kept for 10 years. By enabling Object Lock on the bucket and setting a retention period or legal hold, the bank can make sure those records are not deleted or changed before the required time, even if someone tries to remove them from S3.

Lab Steps

Step 1 – Understand the Object Lock requirement

- Review the purpose of Object Lock.
- Note that it is used to protect objects from deletion or version modification.
- Remember that it is especially useful for compliance and retention requirements.

Step 2 – Review retention modes and legal hold

- Study **Governance mode** and **Compliance mode**.
- Understand that Governance mode can be bypassed with special permissions.
- Understand that Compliance mode cannot be changed or deleted until the retention period ends.

- Review **Legal Hold**, which has no predefined expiration date.

Step 3 – Make sure access and versioning are ready

- Confirm that the required permissions are available to manage Object Lock.
- Ensure that **versioning** is enabled before turning on Object Lock.
- Note that Object Lock cannot be disabled after it is enabled.

Step 4 – Create or edit the bucket

- Open **Amazon S3**.
- Create a new bucket.

Create bucket [info](#)

Buckets are containers for data stored in S3.

General configuration

AWS Region
US East (N. Virginia) us-east-1

Bucket type [info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket namespace
Choose the namespace where you want to create your bucket. [Learn more](#)

☒ **Global namespace**
By default, S3 creates general purpose buckets in the global namespace.

☐ **Account Regional namespace (recommended)**
General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

Bucket name [info](#)

cloudfreeds-object-lock

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

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- Enable Object Lock during bucket creation from the advanced settings, or enable it later from bucket properties.
- Save the changes after acknowledging the warning.

Bucket Key
Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for D3SE-KMS. [Learn more](#)

☐ Disable
☒ Enable

Advanced settings

Object Lock
Store objects using a write-once-read-many (WORM) model to help you prevent objects from being deleted or overwritten for a fixed amount of time or indefinitely. Object Lock works only in versioned buckets. [Learn more](#)

☐ Disable
☒ Enable

Permanently allows objects in this bucket to be locked. Additional Object Lock configuration is required in bucket details after bucket creation to protect objects in this bucket from being deleted or overwritten.

Object Lock works only in versioned buckets. Enabling Object Lock automatically enables Versioning.

Enabling Object Lock will permanently allow objects in this bucket to be locked
After you enable Object Lock for a bucket, you can't disable Object Lock or suspend Versioning for that bucket. [Learn more about Using Object Lock](#)

☒ I acknowledge that enabling Object Lock will permanently allow objects in this bucket to be locked.

After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.

Cancel **Create bucket**

Step 5 – Decide whether to use default retention

- Leave **default retention** disabled if you want to set protection manually per object.
- Enable **default retention** if all objects should share the same lock behavior.
- Choose the retention mode and time period based on your use case.

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Default retention
Automatically protect new objects put into this bucket from being deleted or overwritten.

☐ Disable
☒ Enable

Default retention mode

☒ Governance
Users with specific permissions can overwrite or delete protected object versions during the retention period.

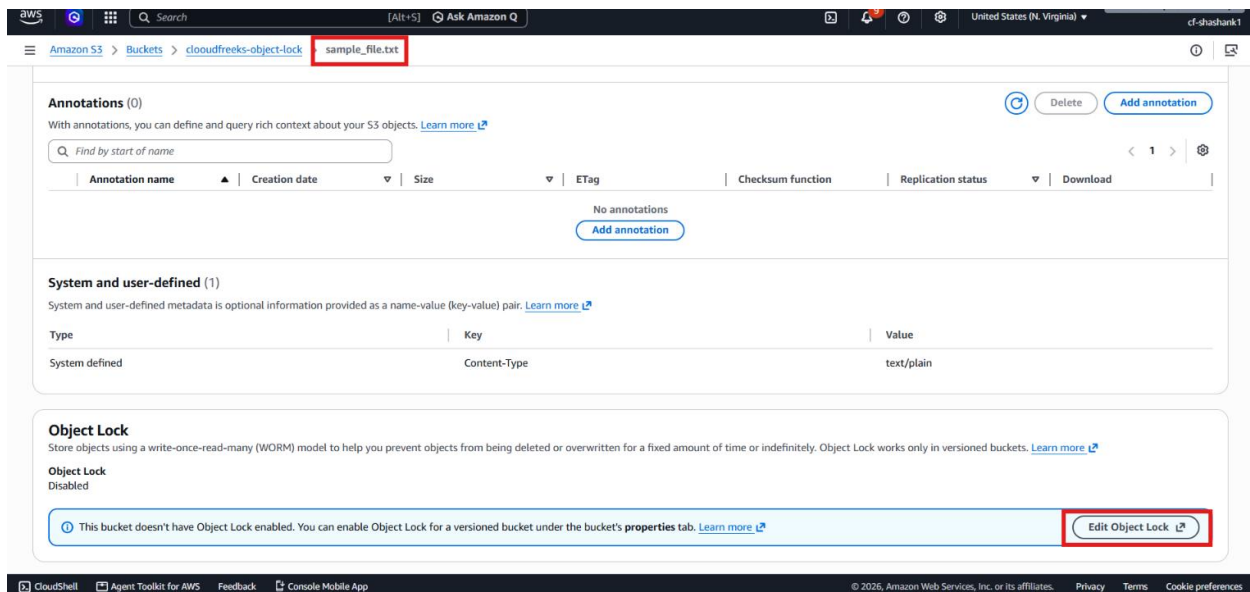
☐ Compliance
No users can overwrite or delete protected object versions during the retention period.

Default retention period
Enter number Days
Must be a positive whole number.

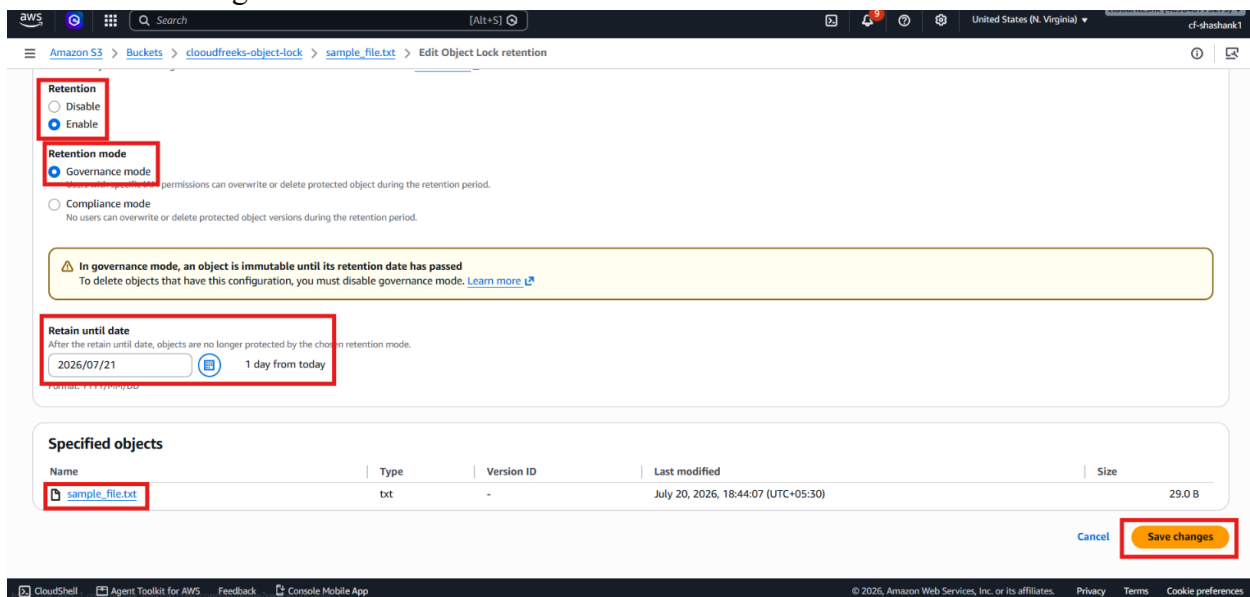
Cancel **Save changes**

Step 6 – Apply object-level retention

- Upload an object to the bucket.
- Open the object's retention settings.



- Choose **Governance mode** or **Compliance mode**.
- Set the retention period for the object.
- Save the changes.



Step 7 – Test the protection

- Try deleting the locked object.
- Confirm that S3 blocks deletion or overwrite while Object Lock is active and you can delete the version but cannot delete the original object.

Failed to delete objects
For more information, see the Error column in the Failed to delete table below. [Diagnose with Amazon Q](#)

Delete objects: status [Close](#)

Deleting objects from a bucket with Object Lock enabled
To use the console to delete objects from a bucket with Object Lock enabled, you must have s3:BypassGovernanceRetention permission. With this permission, specified objects with Object Lock governance mode applied will be deleted. [Learn more](#)

After you navigate away from this page, the following information is no longer available.

Summary

Source s3://cloudfreeds-object-lock	Successfully deleted 0 objects	Failed to delete 1 object, 29.0 B
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[Failed to delete](#) | [Configuration](#)

Failed to delete (1 object, 29.0 B)

Find objects by name

Name	Folder	Version ID	Type	Last modified	Size	Error
sample_file.txt	-	MYXTIP5BwRCpaRidz8ANW9nP0alOm8FO	txt	July 21, 2026, 12:03:55 (UTC+05:30)	29.0 B	Access denied

- If needed, remove the lock or wait for the retention period to end before deletion is allowed.

Step 8 – Try Legal Hold

- Open the object properties.
- Enable **Legal Hold**.

Amazon S3 > Buckets > cloudfreeds-object-lock > sample_file.txt

No tags associated with this resource.

Annotations (0) [Delete](#) [Add annotation](#)

With annotations, you can define and query rich context about your S3 objects. [Learn more](#)

Find by start of name

Annotation name	Creation date	Size	ETag	Checksum function	Replication status	Download
No annotations						

[Add annotation](#)

System and user-defined (1) [Learn more](#)

System and user-defined metadata is optional information provided as a name-value (key-value) pair.

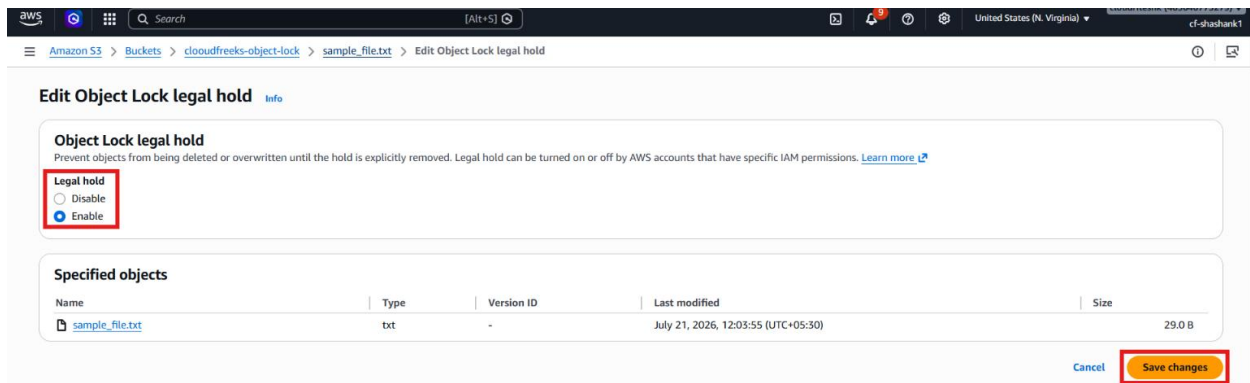
Type	Key	Value
System defined	Content-Type	text/plain

Object Lock legal hold [Edit](#)

Prevents objects from being deleted or overwritten until the hold is explicitly removed. Legal hold can be turned on or off by AWS accounts that have specific IAM permissions. [Learn more](#)

Legal hold
Disabled

- Confirm that the object cannot be deleted by anyone while Legal Hold is active.



- Disable Legal Hold manually when you want deletion or overwrite to be allowed again.

Verification

- Verified that Object Lock protects objects from deletion and modification.
- Verified that **Governance mode** can be bypassed only with special permissions.
- Verified that **Compliance mode** cannot be overridden until the retention period ends.
- Verified that **Versioning** is required before enabling Object Lock.
- Verified that Object Lock cannot be disabled after it is turned on.
- Verified that **Legal Hold** protects objects indefinitely until removed manually.
- Verified that locked objects cannot be deleted or overwritten while protection is active.